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TERM END EXAMINATION

SUMMER - 2024

PROGRAMME: DIPLOMA IN COMPUTER ENGINEERING

COURSE CODE & ITS TITLE : CM5474-LINUX OPERATING SYSTEM

Time Allowed : 03 Hrs

Max. Marks: 70

Instructions:

1. Write your Identity Code Number on question paper.
2. All questions are compulsory.
3. Illustrate your answers with neat sketches wherever necessary.
4. Use of non-programmable calculator is permissible.
5. Figures to the right indicate full marks.
6. Assume suitable additional data, - if necessary and state the assumptions made.
7. Each sub-question in a question carries equal marks unless otherwise specified.
8. Q.1 to Q.6 are as per CO1 to CO6 respectively

Marks

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| Q.1.(A) Attempt any ONE. | 02 |
| a) Write about the term process and its life cycle. | |
| b) List the basic features of linux operating system. | |
| (B) Attempt any TWO. | 08 |
| a) Explain different types of linux shell. | |
| b) Describe basic architecture of linux operating system with neat labelled diagram. | |
| c) Compare Kernel mode with User mode. | |
| Q.2.(A) Attempt any ONE. | 02 |
| a) State the use of cal & date commands. | |
| b) State the function of ordinary file handing command. | |
| (B) Attempt any THREE. | 12 |
| a) Describe grep family commands with example. | |
| b) Demonstrate how to get current file permission and modify it. | |
| c) Differentiate between internal commands and external commands in linux. | |
| d) Demonstrate the use of following linux commands : i) echo ii) passwd iii) mkdir iv) rmdir | |
| Q.3.(A) Attempt any TWO. | 08 |
| a) Explain the syntax of while..do loop by giving example. | |
| b) Write about the wild card characters used in shell and their uses. | |
| c) Explain following commands with example : i) pipe command ii) expr commands | |
| (B) Attempt any TWO. | 08 |
| a) Write a shell script to find greatest & smallest number among given 3 numbers. | |
| b) Design a shell script to find factorial of a number. | |
| c) Design a shell script to find whether entered number is prime or not. | |



Q.4.(A) Attempt any ONE.

02

- a) State the function of startup command.
- b) State any four roles of system administrator.

(B) Attempt any TWO.

08

- a) Demonstrate shutdown operation with commands and activities involved.
- b) Demonstrate adduser & deluser command uses.
- c) Write procedure for system administrator login.

Q.5.(A) Attempt any ONE.

02

- a) State the function of df & du commands.
- b) Enlist run levels in linux with its uses.

(B) Attempt any TWO.

08

- a) Explain the term backup and its need.
- b) Demonstrate disk partitioning in linux with its related commands.
- c) Explain tar command with example.

Q.6. Attempt any ONE.

02

- a) State the use of : i) NIS ii) NFS command.
- b) Write about concept of SAMBA.

(B) Attempt any TWO.

08

- a) Explain the functioning of firewall.
- b) Demonstrate use of following linux commands :
 - i) ping ii) traceroute iii) netstat iv) tcpdump
- c) Design configuration of TCP/IP network address.

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